

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

LECTURE

13

Digestive System, Part - 01

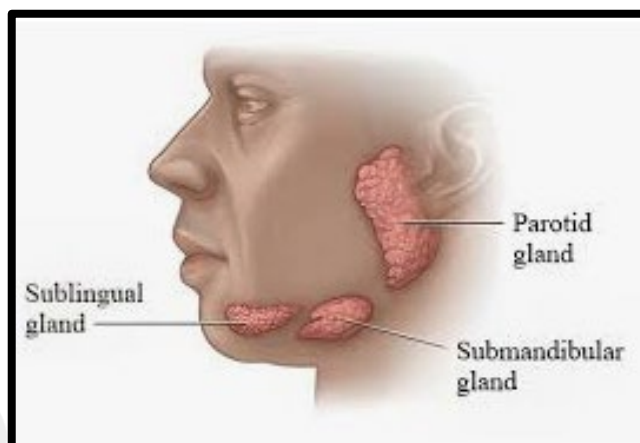
Digestive System, Part - 01

Digestive System

Group of organs responsible for conversion of complex food into its simplest form known as Digestive system.

Digestive System make food into its simplest form:

Complex Food	Simplest Form
Rice/Bread/Potato	Glucose
Pulses/Paneer/Nonveg	Amino acid
Cheese/Butter/Oil	Fatty acid + Glycerol



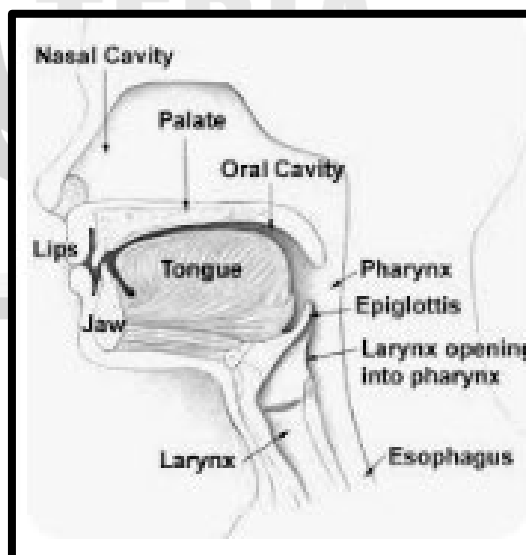
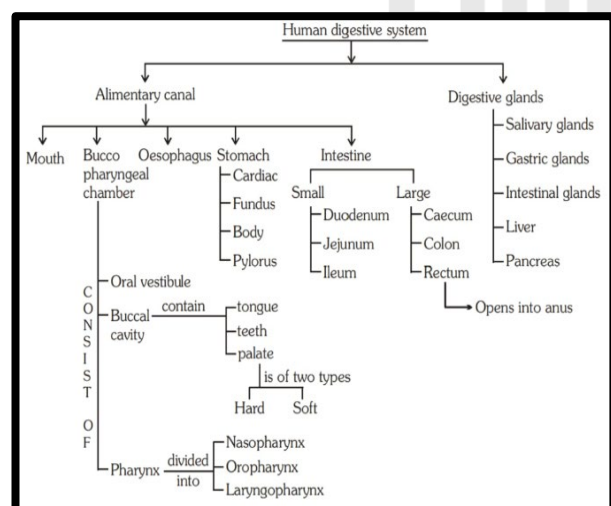
Oral Cavity— Horizontal Transverse slit like operative surrounded by upper and lower lip known as Oral Cavity.

Buccopharyngeal Cavity

- Roof of oral cavity is made up soft and hard potato.
- Floor is made up of muscularis structure called tongue.

Pharynx

The pharynx (commonly known as the throat) is a 12.5-15 cm long muscular tube connecting the nasal/oral cavities to the larynx and esophagus.



Salivary Glands

Salivary glands are exocrine glands that produce and secrete saliva into the oral cavity through ducts to aid in digestion, oral hygiene, and lubrication.

Human Digestive System

- Biomacromolecules (carbohydrates, proteins etc.) in food cannot be utilized by our body in their original form.



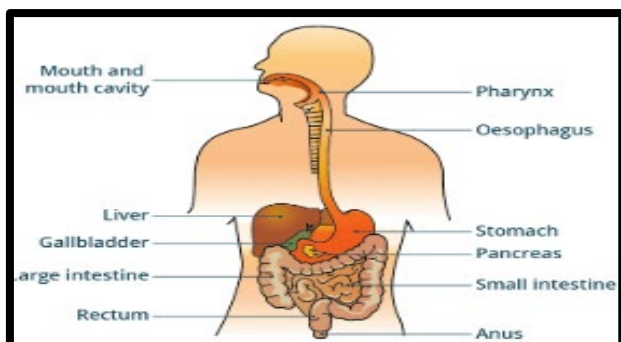


- They have to be broken down and converted into simple substances (glucose, amino acids etc.) in the digestive system.
- During the digestion process, Biomacromolecules like carbohydrates get broken into simple sugars such as glucose, fats into fatty acids and glycerol, proteins into amino acids.

Alimentary Canal

- The food passes through a continuous canal called alimentary canal.
- The canal can be divided into various compartments:
 - (1) the buccal cavity,
 - (2) foodpipe or oesophagus,
 - (3) stomach,
 - (4) small intestine,
 - (5) large intestine ending in the rectum and
 - (6) the anus.
- The activities of the gastro-intestinal tract [alimentary canal] are under neural and hormonal control for proper coordination of different parts.
- The sight, smell and/or the presence of food in the oral cavity can stimulate the secretion of saliva.
- Gastric and intestinal secretions are also, similarly, stimulated by neural signals.
- The muscular activities of different parts of the alimentary canal can also be moderated by neural mechanisms.

Digestive Process



Buccal Cavity or Oral Cavity - Teeth, Tongue, Saliva

- The process of taking food into the body is called ingestion.
- Ingestion happens through mouth.
- The mouth leads to the buccal cavity or oral cavity.
- The oral cavity has a number of teeth and a muscular tongue.
- Each tooth is embedded in a socket of jawbone.
- Majority of mammals including human being forms two sets of teeth during their life, a set of temporary milk or deciduous teeth [milk teeth] replaced by a set of permanent or adult teeth [permanent teeth].
- An adult human has 32 permanent teeth which are of four different types, namely, incisors (I), canine (C), premolars (PM) & molars (M).
- Arrangement of teeth in each half of the upper and lower jaw in the order I, C, PM, M is represented by a dental formula which in human is $\frac{2123}{2123}$ [2-1,1- C,2-PM,3-M]
- The hard chewing surface of the teeth, made up of enamel (Enamel is the hardest substance in the human body and contains the highest percentage of minerals), helps in the mastication (chewing) of food.
- Our mouth has the salivary glands which secrete saliva. The saliva breaks down the starch into maltose.

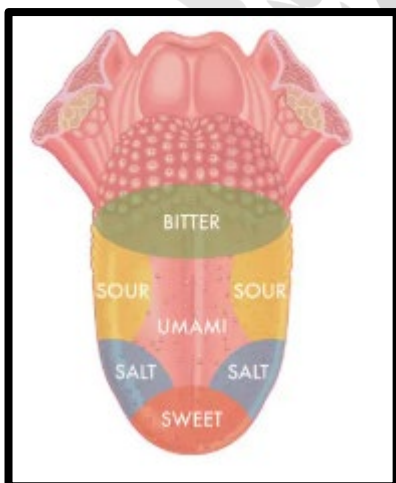




- The saliva secreted into the oral cavity contains electrolytes (Na^+ , K^+ , Cl^- , HCO_3^-) and enzymes, SALIVARY AMYLASE and LYSOZYME.
- The chemical process of digestion is initiated in the oral cavity by the hydrolytic action of the carbohydrate splitting enzyme, the salivary amylase. About 30 per cent of starch is hydrolyzed here by this enzyme (optimum pH 6.8) into a disaccharide – maltose

Tongue

Muscular structure connected with hyoid bone hyoid with the help of frenulum muscles.



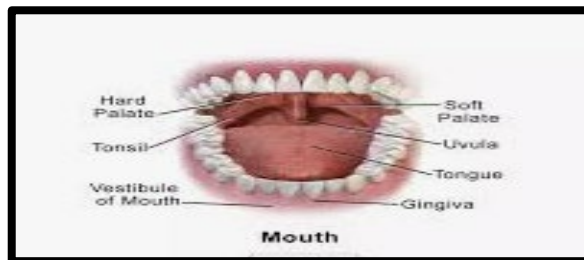
- Extended part of soft palate known as uvula uvulo prevents entry of food into Nasal cavity.
- Cartilaginous flaplike structure known as Epiglottis, Present at the anterior part of larynx. It prevents entry of food into windpipe Trachea.

Four types of teeth present in Oral Cavity—

- **Incisor**—helps in growing. Tusk teeth of elephant is modified incisor.
- **Canine**—Sharp and pointed teeth help in tearing and shredding. More developed in carnivores.
- **Pre molar**— Chewing and grinding large surface area.
- **Molar**— Chewing and grinding large surface area.

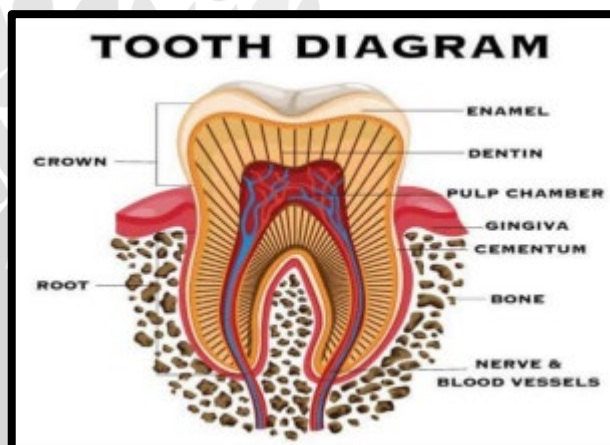
Buccal Cavity

- Thecodont—Present in bony Sacket.
- Heterodont—Different shape & Size.
- Diphyodont—Comes twice in a life.



Tooth

A tooth is a hard, mineralized ectodermal organ embedded in the jawbone, crucial for cutting and chewing food (digestion), supporting facial structure, and assisting in speech.



Stomach

- The inner lining of the stomach secretes mucous, hydrochloric acid and digestive juices.
- The mucous protects the lining of the stomach.
- The acid kills many bacteria that enter along with the food and makes the medium in the stomach acidic.
- The digestive juices break down the proteins into simpler substances.
- A muscular sphincter (gastro-esophageal) [a ring of muscle surrounding and serving to guard or close an opening] regulates the opening of esophagus into the stomach.
- The stomach, located in the upper left portion of the abdominal cavity, has three major parts – a cardiac portion into which the esophagus opens, a fundic region and a pyloric portion which opens into the first part of small intestine



**Small intestine**

- Small intestine is distinguishable into three regions, a 'C' shaped duodenum, long coiled middle portion jejunum and a highly coiled ileum.
- The opening of the stomach into the duodenum is guarded by the pyloric sphincter. Ileum opens into the large intestine.
- The small intestine is highly coiled and is about 5 meters long. It receives secretions from the liver and the pancreas. Besides, its wall also secretes juices.
- The digested food passes into the blood vessels in the wall of the intestine. This process is called absorption.
- The inner walls of the small intestine have thousands of finger-like outgrowths. These are called villi (singular villus).
- The villi increase the surface area for absorption of the digested food.
- Villi are supplied with a network of capillaries and a large lymph (a colourless fluid containing white blood cells) vessel called the lacteal.
- The absorbed substances are transported via the blood vessels to different organs of the body where they are used to build complex substances such as the proteins required by the body.
- This is called assimilation.
- In the cells, glucose breaks down with the help of oxygen into carbon dioxide and water, and energy is released.
- The food that remains undigested and unabsorbed then enters into the large intestine.

The Summary of Absorption in Different Parts of Digestive System

Mouth	Stomach	Small Intestine	Large Intestine
Certain drugs coming in contact with the mucosa of mouth and lower side of the tongue are absorbed into the blood capillaries lining them.	Absorption of water, simple sugars, and alcohol etc. takes place.	Principal organ for absorption of nutrients. The digestion is completed here and the final products of digestion such as glucose, fructose, fatty acids, glycerol and amino acids are absorbed through the mucosa into the blood stream and lymph.	Absorption of water, some minerals and drugs takes place.

Large intestine

- The large intestine is wider and shorter than small intestine.
- It is about 1.5 metre in length.

- Its function is to absorb water and some salts from the undigested food material.
- The remaining waste passes into the rectum and remains there as semi-solid faeces.
- The faecal matter is removed through the anus from time-to-time. This is called egestion.
- Ingestion → Digestion → Absorption → Assimilation Egestion
- It consists of caecum, colon and rectum. Caecum is a small blind sac which hosts some symbiotic micro-organisms.
- A narrow finger-like tubular projection, the vermiform appendix which is a vestigial organ [small remnant of something that was once more noticeable], arises from the caecum.
- Appendix was helpful in digesting roughage (fibrous indigestible material in vegetable foodstuffs which aids the passage of food and waste products through the gut).
- Thousands of years ago, when man used to eat roots, leaves, etc., it was essential.
- But now it has lost its significance.
- The caecum opens into the colon.
- The colon is divided into three parts – an ascending, a transverse and a descending part.
- The descending part opens into the rectum which opens out through the anus.
- No significant digestive activity occurs in the large intestine.
- The functions of large intestine are: absorption of some water, minerals and certain drugs; secretion of mucus which helps in adhering the waste (undigested) particles together and lubricating it for an easy passage.
- The undigested, unabsorbed substances called faeces enters into the caecum of the large intestine through ileo-caecal valve, which prevents the back flow of the faecal matter.
- It is temporarily stored in the rectum till defecation.

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